Fakultät Informatik

From CAD Software to Slicer

Workflow Optimization using Print Hints in 3MF Files for 3D Printing

Bachelorarbeit im Studiengang Medieninformatik

vorgelegt von

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Kurzdarstellung

Abstract

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Chapter 1.

Introduction

"*makers*, a community of enthusiasts who focus on fabrication, invention and experience sharing, and who collaborate and learn in environments known as *makerspaces*"[6, p. 384]

"thousands of public institutions, such as schools, libraries, and universities are now setting up print centers or creative hubs for end users to make use of digital fabrication technologies"[6, p. 384]

"makerspaces that are often run by enthusiasts [. . .] attract engineers, entrepreneurs, inventors, hackers, craftsmen, and artists"[6, p. 384]

"casual makers serve as a proxy for what it will be like for the general public to use 3D printing"[6, p. 384]

"if HCI is going to be at the forefront of inventing new fabrication design tools and interfaces[. . .], we need to look beyong enthusiasts and understand the challenges and opportunities that exist in facilitating the interactions of casual makers with fabrication technology"[6, p. 384]

"current tools that focus on supporting only one aspect of the workflow (*e. g.* , only modelling or only printing) may be less useful for casual makers and there is opportunity in HCI research to innocate on design tools that take into account the interdependencies within the 3D printing workflow"[6, p. 385]

"while lowering the usability barriers of fabrication tools is important, it is perhaps even more important to weave-in expertise on best practices and troubleshooting tips within the walk-up-and-use tools used by casual makers to reduce theiir dependecy on operators"[6, p. 385]

"opportunities that exist in building the next generation of inter-connected fabrication tools with features for supporting expertise sharing"[6, p. 385]

"enthusiasts and professional users—that is, they were trained or technically literate in fabrication"[6, p. 386]

Definition of slicer

“Processing the 3D model occurred in specialized software known as a slicer (as it slices the model into layers for the printer), which allowed users to select print settings, such as layer height and print speed”[6, p. 389].

“If the user’s print failed or did not match their design intent, they would have to adjust either their model or print settings to correct the issue, often with the operator’s assistance”[6, p. 389]

Definition of supports

“With FDM printers, geometric constraints relating to overhangs and base size can be partially addressed using automatically generated scaffolding known as supports and rafts”[6, p. 391]

“many print centers enforced final check-overs by the operators on users’ designs and print settings before printing to reduce failed prints”[6, p. 392]

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Glossary

library A suite of reusable code inside of a programming language for software development. i

shell Terminal of a Linux/Unix system for entering commands. i